

A Minimalistic Quadrotor Navigation Strategy for Indoor Multi-floor Scenarios

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Abstract This paper presents the application of a minimalistic navigation strategy, based on the well-known BUG2 algorithm, to solve the problem of reaching a goal position in a multi-floor indoor scenario using a quadrotor. Examples of this scenario include buildings and in general cluttered indoor areas. As far as energy backup is concerned the quadrotor shows strict constraints: for this reason implementing a low-consumption navigation strategy is a major issue. We present a two-layer navigation strategy, called MF-BUG2, useful to navigate in multi-floor buildings starting from the ground floor toward the last or vice versa while searching for an interesting physical quantity (i.e., gas leak, electromagnetic source). In the lower layer a BUG-like algorithm is able to drive the flying robot, equipped with a salient-cue sensor and a laser-range-finder, toward the estimated position of goal on the horizontal plane while avoiding obstacles and using minimal computational power and memory (the boundary-following behavior uses an Artificial Potential Field to navigate around the obstacles). If the estimated goal position is reached but the salient-cue-sensor does not detect a salient quantity the higher level of the planner calls Dijkstra algorithm to compute the minimum-distance path to change the floor, assuming to know in advance the 2D position of the passages among different floors, and then moves vertically. The overall strategy is useful for *indoor inspection* in hazardous scenarios. The algorithm is validated in simulation, investigating the robustness with respect to the laser-range-finder noise.

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1 Introduction

Research in the field of flying robots has shown a great potential, during the past decade, thanks to the extensive use of quadrotors [1]. The mechanical simplicity of this machine has encouraged a lot of researchers to develop autonomous or semi-autonomous flying robots able to navigate in outdoor and indoor scenario [2].

The applications of this technology ranges from agriculture to monitoring, from photogrammetry to Urban Search and Rescue (USAR). The main drawback lies in the small ability of carrying energy backup. This implies two main constraints:

1. on the sensor payload, due to the limited lifting thrust;
2. on the computational power, due to the limited current/hour capacity;

As a consequence, it is necessary to design navigation systems that uses minimal information and sensing in order to guarantee a low energy consumption. Surprisingly enough, the majority of the systems presented in the literature adopt an approach in designing quadrotor platforms that do not comply with constraints (1) and (2) above. Since the mainstream approach to navigation involves SLAM (Simultaneous Localization and Mapping) as a core part, heavy computational algorithms are needed to perform signal processing, features extraction, map building, and path planning [3]. The nature of the *representation* of the world plays a fundamental role in this issue. A *global approach* provides the robot with the “most accurate” representation of the workspace considering the presence of incomplete data and sensor noise [4], while a *local approach* implements reactive, sensor-driven, navigation behaviors that takes inspiration from biological beings [5–7]. It is worth noting that, referring to this framework, SLAM-based approaches can be considered *hybrid* because of use of a local information to build a global representation. A global approach is definitely not appropriate for our application, while it seems reasonable to adopt a hybrid approach built upon a local navigation strategy that satisfies technological constraints we are considering and allows to optimize the robot in terms of energy backup and miniaturization.

This paper presents the design of a two-layer hybrid navigation system for a quadrotor, called MF-BUG2, used to compute *online* and follow a path between a start and goal position in a multi-floor indoor scenario. The goal must be associated with some salient cue (visual, electromagnetic, olfactory [8]). The *lower layer* is based on the well-known BUG2 algorithm that implements a local navigation approach, as discussed before, and is responsible for navigating in the horizontal plane. The *higher layer*, based on the Dijkstra algorithm, computes a sequence of subgoals (which are fed to the BUG2 algorithm in the lower layer) using the global information of a graph induced by the set W of 2D positions of the passages among the floors and is responsible for vertical navigation. Thereafter, assuming to navigate in a multi-floor building, the navigation system requires just the 2D position of the

goal, the 2D position of the robot, and the set W defined above. We assume to have a small quadrotor equipped with a 2D laser-range finder and a sensor able to sense the cue. The assumption about *indoor localization* is realistic thanks to UWB localization technology [9].

The work described is part of the research project PRISMA,¹ aimed at intervention in post-disaster scenarios, and is useful for *indoor inspection*, seeking for gas leaks on vertical pipes in multifloor buildings or other similar tasks.

The paper is organized as follows. Section 2 presents the literature, focusing on the attempt to extend the minimal sensing problem to three-dimensional environments. Section 3 describes the navigation system, while Sect. 4 shows the results obtained in simulation. Conclusions follow.

2 Related Work

Navigation of robots with minimal information has been explored preeminently in 2D scenarios [10, 11]. The research community has been focused on extending BUG-like algorithms [6, 12–14]. This class of navigation algorithms, thoughts for 2D point automata, and based on two possible behaviors (i.e., motion-to-goal, boundary-following) is huge and different classifications are present in the literature [4]. BUG2 needs to know the position of the automaton, the position of the goal (from which the distance is computed) and the information acquired from a contact sensor. An extension is TangentBug [15], which uses a range-sensor with a field-of-view of 360° to acquire the distances from the obstacles around the robot and stores a set of *continuity intervals* in a data structure called Local Tangent Graph (LTG). Wedgebug algorithm [16] (and its extension RoverBug [16]) assumes to detect distances in a wedge. Ibug [17] uses an intensity sensor, which measures the signal strength emanating from the goal.

The first approach for solving the path planning problem for a mobile robot operating in an unknown three-dimensional environment with minimal information was presented in [18, 19]. The authors discussed a crucial link between the path planning problem and the problem of visually exploring a three-dimensional environment, concluding that visual exploration is necessary for successfully planning the motion of a robot in unknown three-dimensional environments and introducing the notion of *exploratory algorithms*. Following this intuition Kamon and colleagues introduced 3DBug [20]. The robot is equipped with an ideal spherical range sensor with infinite detection range and the algorithm uses two modes of motion: motion-toward-the-target and obstacle-surface-transversal (that is a generalization of boundary-following for 3D polytopical obstacles). The key point of the algorithm is the capability of expanding the knowledge of the obstacle, using range data, while attempting to reach an exit point along an obstacle surface. Due to utilization of an ideal spherical sensor, 3DBug is an interesting algorithm from a theoretical perspective. However, it is really hard to implement for its assumptions.

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A review of the literature shows that the issue of navigating in a realistic 3D scenario with minimal sensing has not been successfully addressed without relaxing the energy constraint and assuming ideal sensors [3]. We start focusing our effort, following the purposes of project PRISMA, on the case of a multi-floor scenario.

3 Problem Statement and System Architecture

After an earthquake or other calamities there is the need to inspect indoor buildings, checking for structural integrity, damages, unsafe situation, or victims. The use of robots allows to automate inspection tasks, helping rescuers in hazardous areas. Depending on the nature of the task the robot has to be equipped with a salient-cue-sensor to perceive an interesting physical quantity (e.g., gas, light). In this cluttered scenarios it is unreasonable to know in advance shape, dimensions, and position of obstacles (e.g., walls, rubbles) while it is possible, looking at the planimetry, to estimate the position of vertical passages among different floors (e.g., stairwells) that could be eventually occluded. Thanks to research in the field of indoor localization it is today possible to localize a robot, also in cluttered areas, using different systems among which the most promising is based on Ultra Wide Band technologies [9]. Due to the tridimensionality of the considered scenario a flying robot has a lot of advantages with respect to ground vehicles, and a quadrotor is the most versatile platform. A quadrotor [21] is a rotorcraft that exploits the lift force provided by four rotors usually mounted in cross configuration. In order to define an orientation (or attitude) around its center of mass, three dynamic parameters, namely the angles of yaw, pitch, and roll are usually defined. The overall force F_B and torque M_B produced by the propellers, expressed in the body frame B, are used to control the quadrotor causing it to pitch, roll, or yaw. By changing these three angles we are able to make the quadrotor maneuver in almost any direction. The quadrotor can move in the XY plane, it can change its altitude along the Z axis, and can change its heading angle Φ (i.e., holonomic on the XY plane). Due to the fact that the quadrotor has no contact with the ground we do not use the Φ angle in our representation.

In order to formulate the algorithm let us introduce a number of definition. Let us define $S = (x_S, y_S) \in \mathbb{R}^2$ and $G = (x_G, y_G) \in \mathbb{R}^2$ as the Start and Goal locations of a mission, whereas $C = (x_C, y_C)$ is the current robot position. Let us define an *m-line* as the straight line connecting S and G , and W as the vector of all the positions of the passages between floors, where each positions w_i , called waypoint, is expressed as $w_i = (x_i, y_i)$. Let us define a totally connected weighted graph $P = (N, A, M)$, with $N = W \cup G$ the initial set of nodes, A the set of bidirectional arcs, and a set of weights M containing the Euclidian distances between nodes. Finally, let us define $d\langle X, Y \rangle$ as a function which measures the Euclidian distance between any two points $X, Y \in \mathbb{R}^2$. We suppose to equip the quadrotor with two sensors: a laser-range-finder with a field of view of 360° and a salient-cue sensor SC triggered by some interesting quantity as specified above.

Algorithm 1: MF- BUG2

Input: A vector W of waypoint coordinates between floors
 $W = ((x_1, y_1), (x_2, y_2), \dots, (x_n, y_n))$
Input: The start and goal position S and G
Input: The salient-cue sensor SC

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1   $j \leftarrow 1$ 
2   $L_o \leftarrow S$ 
3  moving-vertically = 0
4  while  $S \neq G$  do
5      motion-to-goal
6      Update  $C$ 
7      if Goal is reached  $\wedge SC = 1$  then Exit
8      if Goal is reached  $\wedge SC = 0 \wedge$  moving-vertically = 1 then
9          Move along Z axis
10          $G \leftarrow MainGoal$ 
11         moving-vertically  $\leftarrow 0$ 
12     if Goal is reached  $\wedge SC = 0$  then
13         ShortestPathWaypoint  $\leftarrow Dijkstra(\langle C, W, G \rangle, A, M)$ 
14          $MainGoal \leftarrow G$ 
15          $G \leftarrow ShortestPathWaypoint$ 
16         moving-vertically  $\leftarrow 1$ 
17     if Obstacle is encountered then  $H_j = C$ 
18         boundary-following
19         while  $C \neq H_j$  do
20             if  $C = G$  then Exit
21             if  $C$  is on the  $m$ -line then
22                 if  $d\langle C, T \rangle < d\langle H_j, T \rangle$  then
23                     if the robot does not cross an obstacle at  $C$  then
24                          $L_j = C$ 
25                          $j \leftarrow j + 1$ 
26                     Break
27
28 
```

As mentioned above MF-BUG2 is a two-layer navigation system. The lower level exhibits three different behaviors: motion-to-goal, boundary-following, moving-vertically. The laser scanner is used to emulate a contact sensor, yet keeping the quadrotor at a safety distance from obstacles. During motion-to-goal (line 5), which is the starting behavior, the robot moves along the m -line, until an obstacle. At this point the algorithm switches to boundary following (line 18) behaviour and stores the Cartesian position of the *hit-point* H_j and the distance $d\langle H_j, G \rangle$. It keeps following the boundary of the obstacle until it cross again the m -line (lines 21–23) and the distance between the current position of the robot and the goal $d\langle C, G \rangle$ is smaller than $d\langle H_j, G \rangle$. At that point, a *leave-point* L_j is defined and the motion-to-goal behavior is invoked again. The planner, as usual for the BUG2, tries to minimize the distance between the robot and the goal.

As soon as the estimated goal position is reached on the horizontal plane, if the salient-cue sensor is not triggered, the higher level of the navigation system invokes $Dijkstra(\langle C, W, G \rangle, A, M)$ to plan the shortest path on the induced weighted directed graph $P = (N, A, M)$ with an extended set of nodes $N = \langle C, W, G \rangle$ and edges A . The node C , representing the current position, is now connected to each node of W with simple directed arcs and each node of W is connected to G with a simple directed arc. Note that there is no arc between C and G . The Dijkstra algorithm returns the shortest path between C and G , composed of three nodes where the second node is the 2D position of *A vertical passage* (waypoint), and then is set as a subgoal for BUG2 (line 11, 12), while the main goal position is stored in the variable *MainGoal*. As soon as this latter is reached the robot moves along Z of a constant altitude (moving-vertically behavior) changing the floor while using an obstacle avoidance strategy and finally start again moving toward the main goal (line 9). The overall navigation behavior makes the robot moving from a floor to the next one, reaching the estimated goal position at each floor, until it find the cue and then terminates. The proposed algorithm is intuitively $O(n)$ in space complexity where $n = N - 1$ and N is the number of floor of the building.

The boundary following behavior is implemented using an Artificial Potential Field, as used in *μnav* [4]. The robot motion law is given by:

$$v = [W_g(U)g + W_t(U)t + W_f(U)f]V_{ref} \quad (1)$$

where v is the velocity of the robot, V_{ref} is a reference value for the speed, g is a unit vector directed to the goal, t is a unit vector tangent to the APF equipotential line, f is a unit vector directed along the inverse gradient of the potential field, U is the current value of the potential field, and W_g , W_t , W_f are weights which depends on U .

4 Simulation Results

The simulations are performed in VREP² (Virtual Robot Experimentation Platform) that provides an integrate environment with a dynamic engine, 3D objects (robot, furnitures) and sensors. VREP allows to achieve a realistic physical simulation, including collisions with the environment and dynamic flight simulation of the rotorcraft. The navigation system is implemented from scratch in ROS,³ using the Point Cloud Library⁴ (PCL) to acquire the laser scanner data.

Based on the model of a simple quadrotor, equipped with a Hokuyo URG-04LX laser-scanner and a radio-transceiver (with 0.5 m of range) as salient-cue sensor (Fig. 1a, b), the validity of the proposed algorithm has been tested in a two-floors

²<http://www.coppeliarobotics.com>.

³<http://www.ros.org>.

⁴<http://www.pointclouds.org>.

Table 1 H = num of hitpoints, L = num of leave points, U = num of floors passages, Lenght = path's length (meters), RD = Residual Distance to the goal (meters), Time = travel time

Simulation	H	L	U	Lenght	RD	Time
1	0	0	1	14.74	0.29	1:29
2	2	2	1	22.50	0.26	2:14
3	2	1	1	17.39	0.29	1:29
4	4	4	1	16.62	0.28	2:17
5	2	1	1	22.08	0.29	2:11
6	2	2	1	24.11	0.24	3:44
7	3	3	1	24.75	0.27	4:32
8	3	3	1	40.21	0.29	4:46
9	2	2	1	19.00	0.27	2:35
10	5	5	1	23.06	0.28	1:56

These measurements are taken without adding noise ($\sigma = 0$)

Table 2 Measurements relative to simulation 9 of Table 1 adding Gaussian Noise

Simulation	σ	Lenght	Time
1	0.025	18.79	2:37
2	0.05	18.72	2:24
3	0.1	13.61	2:39

σ = Standard Deviation of AWGN noise with $\mu = 0$, Lenght = path's lenght (meters), Time = travel time

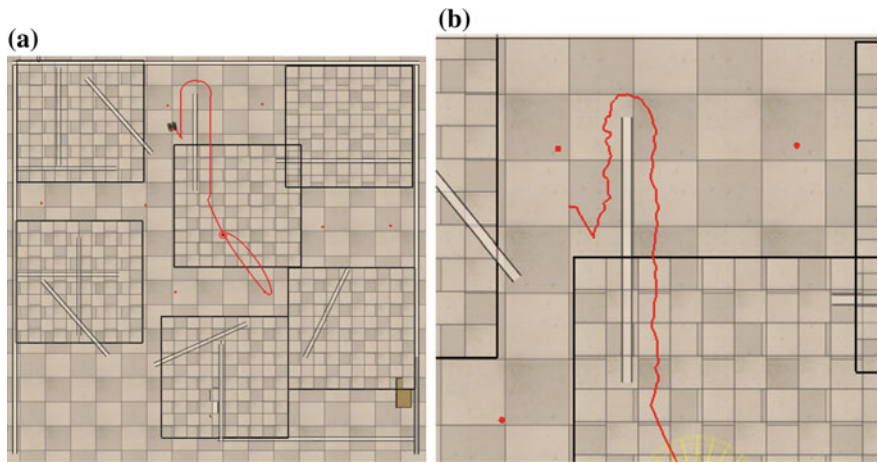


Fig. 2 **a** The red line is the path from start to goal. The yellow circle is the sensing radius of the salient-cue sensor. **b** Noisy trajectory with $\sigma = 0.1$ and $\mu = 0$

with an increasing Standard Deviation of $\sigma = 0.025, 0.05, 0.1$. With $\sigma = 0.025$ and $\sigma = 0.05$ we do not have a significant change in travel-time and length, while with $\sigma = 0.1$ is observable a shortening of the path length due to the fact that the robot navigates closer to the obstacle.

5 Conclusions

The article presented a simple approach to plan the motion of a quadrotor in a indoor multi-floor scenario with minimal information. The navigation system is explicitly designed for autonomous inspection tasks in hazardous multi-floor buildings, assuming the availability of a wireless localization systems based on Ultra Wide Band⁵ technology. Our design objective is to satisfy the energy constraint typical of quadrotors. The navigation system is based on the well-known minimalistic BUG2 and Dijkstra algorithms. The robot requires only a laser-range-finder and a salient-cue sensor to capture interesting quantities for post-disaster indoor inspection. Experiments in simulation have shown that the proposed planner performs well, also in presence of noisy measurements, and can be used in real scenarios. Our future effort will be focused on implementing and testing the strategy using a real quadrotor.

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⁵<http://disal.epfl.ch/page-47308-en.html>.

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